

Statistical Shape Analysis

Short Course: Based on Dryden and Mardia (2016)

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Outline

- 1 Motivation and Data Structures
- 2 Removing Translation and Measuring Size
- 3 Shape Spaces and Geometry
- 4 Procrustes Analysis
- 5 Tangent Coordinates and Multivariate Statistics
- 6 Inference and Group Comparisons
- 7 Outlines, Curves, Surfaces, and Unlabelled Data
- 8 Modern Directions
- 9 Summary and Discussion

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Learning Goals

By the end of this mini-course, students should be able to:

- ① define size, form, and shape for landmark configurations;
- ② construct centred, scaled, and registered configurations;
- ③ derive the ordinary Procrustes fit using the singular value decomposition;
- ④ explain the pre-shape sphere and Kendall shape space;
- ⑤ compute and interpret Procrustes means, tangent coordinates, PCA, and regression;
- ⑥ describe some current developments in statistical shape analysis.

Target audience

Graduate students, postdocs, other researchers with linear algebra, multivariate statistics, and some exposure to R.

Notation

- k : number of landmarks.
- m : ambient dimension, usually $m = 2$ or $m = 3$.
- n : number of specimens.
- $X_i \in \mathbb{R}^{k \times m}$: configuration matrix for specimen i .
- $\mathbf{1}_k$: k -vector of ones.
- $SO(m) = \{R : R^T R = I_m, \det(R) = 1\}$: rotation group.
- $\|A\|_F = \sqrt{\text{tr}(A^T A)}$: Frobenius (or Euclidean) norm.

We treat rows as landmarks ($i = 1, \dots, k$) and columns as coordinates (e.g. x,y,z).

Why shape analysis?

Objects are everywhere: skulls, bones, cells, proteins, handwritten digits, organs, surfaces, and geographic point patterns.

Scientific questions often have the form:

- Are two populations different in average shape?
- What are the dominant modes of shape variation?
- Does shape change with size, age, disease, or environment?
- Can shape predict group membership or functional outcome?

The difficulty: shape is not just a vector in ordinary Euclidean space.

Definition: Shape

Definition (Shape)

Shape is all geometrical information that remains when location, scale, and rotational effects are filtered out from an object.

For configurations X and Y , the same shape means

$$Y = \beta X R + \mathbf{1}_k \gamma^T, \quad \beta > 0, \quad R \in SO(m), \quad \gamma \in \mathbb{R}^m.$$

Core idea

Statistical shape analysis is statistics on equivalence classes under similarity transformations.

Size, Shape, and Form

Shape

Remove translation, rotation, and scale.

$$X \sim \beta XR + \mathbf{1}\gamma^T.$$

Size-and-shape / Form

Remove translation and rotation, but retain scale.

$$X \sim XR + \mathbf{1}\gamma^T.$$

Practical warning

If biological size is meaningful, do not remove it without asking whether allometry is part of the scientific question.

Landmark Data

A landmark is a point of correspondence on each object.

Type	Example
Anatomical	skull sutures, eye corners, bone junctions
Mathematical	high-curvature points, extremal points
Pseudo-landmarks	equally spaced points along a curve
Semi-landmarks	points allowed to slide tangentially

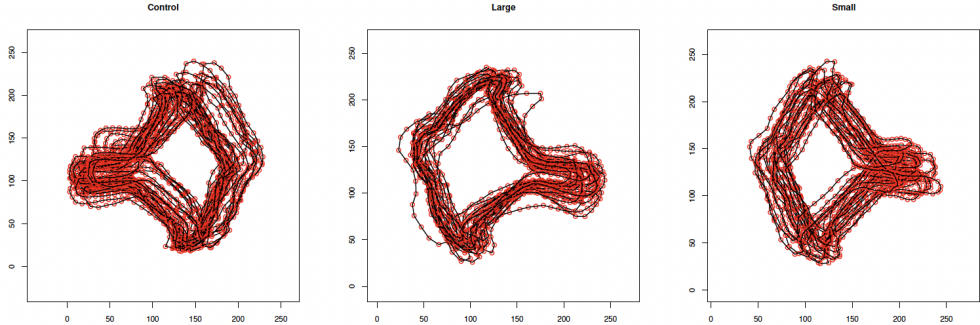
The more artificial the landmark, the more carefully correspondence must be justified.

Examples

- Mouse vertebrae: selected large, small, and control groups.
- Macaque skulls: 3D anatomical landmarks, sex differences.
- Ape skulls: species and sex comparisons, face vs braincase.
- Schizophrenia MR images: brain landmarks in two groups.
- Sand grains and microfossils: outline data via pseudo-landmarks.
- Proteins: 3D active-site matching with uncertain correspondences.

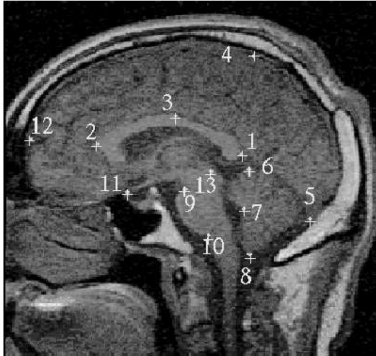
The same geometry recurs across very different scientific domains.

Biological morphometrics



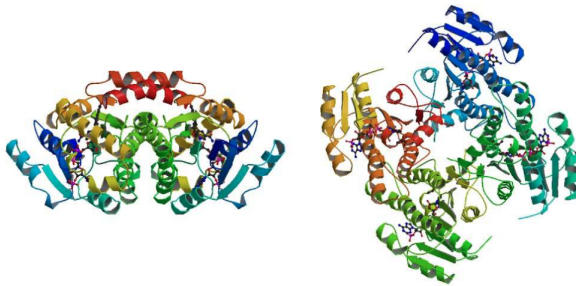
Morphometrics studies biological form quantitatively. Landmark-based methods allow researchers to compare anatomical structures across species, developmental stages or environmental conditions.

Medical imaging applications



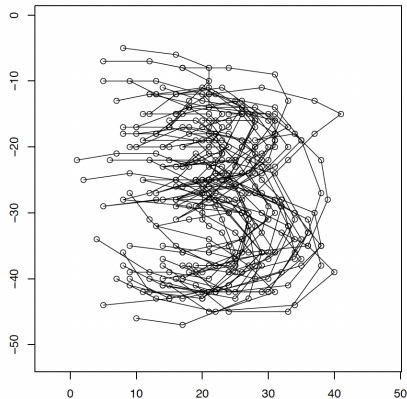
Shape analysis is widely used in neuroimaging and radiology. Examples include studying cortical thickness, hippocampal deformation and anatomical changes associated with neurological disorders.

Protein structure applications



Protein structures can be compared geometrically to identify functional similarities and evolutionary relationships. Shape matching methods are important in structural biology and drug discovery.

Computer vision applications



In computer vision, shape descriptors are used for object recognition, handwriting analysis and image retrieval. Statistical models help distinguish meaningful variation from noise.

Mini-Exercise 1

For your own research area, identify:

- ① the object whose shape matters;
- ② possible landmarks or curves;
- ③ nuisance transformations;
- ④ a scientific covariate of interest;
- ⑤ a possible source of correspondence error.

Discussion

A shape-analysis study starts before statistics: the landmarking protocol is part of the model.

Configuration Matrix

For k landmarks in m dimensions (usually $m = 2$ or $m = 3$),

$$X = \begin{bmatrix} x_{11} & \cdots & x_{1m} \\ \vdots & \ddots & \vdots \\ x_{k1} & \cdots & x_{km} \end{bmatrix} \in \mathbb{R}^{k \times m}.$$

Rows are labelled landmarks. A sample is

$$X_1, \dots, X_n.$$

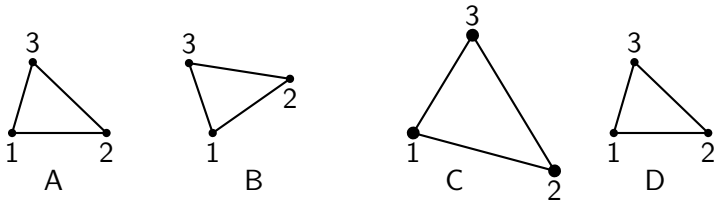
For 2D data, it is often convenient to use complex coordinates:

$$z_j = x_j + iy_j, \quad z = (z_1, \dots, z_k)^T \in \mathbb{C}^k.$$

A First Picture: Similar Triangles

A, B, C: same labelled shape

D: same triangle geometry only if labels are ignored



Similarity transformations preserve labelled shape; permutations create the unlabelled shape problem.

Labelled vs Unlabelled

Labelled landmarks

Landmark j on one object corresponds to landmark j on every other object.

Unlabelled landmarks

Correspondence is unknown or irrelevant; permutations of labels are equivalent.

Unlabelled registration typically requires

$$\min_{P \in \mathcal{P}, R \in SO(m), \beta > 0, \gamma} \|X - \beta P Y R - \mathbf{1} \gamma^T\|_F^2,$$

where P is a permutation matrix.

This is both statistical and combinatorial.

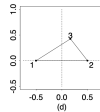
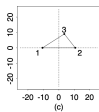
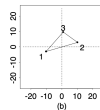
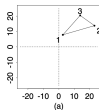
Simple shape co-ordinates

Bookstein coordinates $(u_j^B, v_j^B)^T, j = 3, \dots, k$, are the remaining coordinates of an object after translating, rotating and rescaling the baseline to $(-1/2, 0)$ and $(1/2, 0)$ so that

$$u_j^B = \{(x_2 - x_1)(x_j - x_1) + (y_2 - y_1)(y_j - y_1)\} / D_{12}^2 - 1/2,$$

$$v_j^B = \{(x_2 - x_1)(y_j - y_1) - (y_2 - y_1)(x_j - x_1)\} / D_{12}^2,$$

where $j = 3, \dots, k$, $D_{12}^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2 > 0$ and $-\infty < u_j^B, v_j^B < \infty$.



Simple shape co-ordinates

```
data(mice)
qset2<-mice$x[, ,mice$group=="s"]
u<-bookstein2d(qset2)$bshpv
plotshapes(u,symbol=as.character(1:6))
```

For example, the first mouse has Bookstein shape variables given by the last four rows of:

```
> u[, ,1]
, , 1
```

	[,1]	[,2]
[1,]	-0.50000000	0.00000000
[2,]	0.50000000	0.00000000
[3,]	0.09121051	0.2826022
[4,]	0.04065577	0.5089238
[5,]	-0.07367575	0.2735386
[6,]	-0.02672182	-0.3054334

Statistical Tasks

Common tasks:

- 1 Estimate a mean shape.
- 2 Quantify shape variability.
- 3 Compare groups.
- 4 Relate shape to covariates.
- 5 Classify using shape.
- 6 Visualize deformation modes.

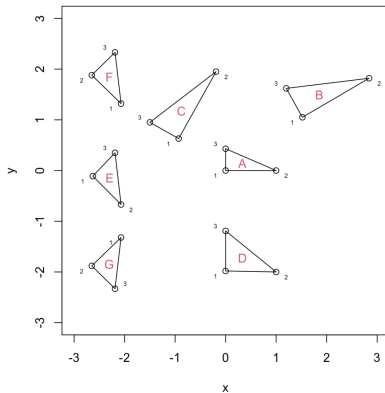
The central challenge is to separate nuisance transformations from scientifically meaningful geometry.

Mini-Exercise 2

Which of the triangles below have approximately the same size and shape? (assume by 'shape' we mean labelled shape).

Which triangles have approximately the same shape but different sizes?

Which triangles have different shapes but approximately the same 'reflection shape'? i.e. a translation, rotation, scale and a reflection will match them exactly.



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Translation Removal

The centroid of configuration X is

$$\bar{x} = \frac{1}{k} X^T \mathbf{1}_k \in \mathbb{R}^m.$$

Define the centring matrix

$$C = I_k - \frac{1}{k} \mathbf{1}_k \mathbf{1}_k^T.$$

The centred configuration is

$$X_c = CX = X - \mathbf{1}_k \bar{x}^T.$$

Properties:

$$C^T = C, \quad C^2 = C, \quad C \mathbf{1}_k = 0.$$

Translation Invariance

For any translation vector $\gamma \in \mathbb{R}^m$,

$$C(X + \mathbf{1}_k \gamma^T) = CX + C\mathbf{1}_k \gamma^T = CX.$$

Thus centring removes location exactly.

The centred configurations live in the linear subspace

$$\{X \in \mathbb{R}^{k \times m} : \mathbf{1}_k^T X = 0\},$$

which has dimension

$$m(k - 1).$$

Centroid Size

The standard size measure is centroid size:

$$S(X) = \|CX\|_F = \left(\sum_{j=1}^k \|x_j - \bar{x}\|^2 \right)^{1/2}.$$

Equivalent expression:

$$S^2(X) = \frac{1}{k} \sum_{1 \leq a < b \leq k} \sum_{1 \leq a < b \leq k} \|x_a - x_b\|^2.$$

Centroid size is invariant under translation and rotation, and scales linearly:

$$S(\beta X) = \beta S(X), \quad \beta > 0.$$

Proof of the Pairwise-Distance Formula

Let $x_1, \dots, x_k \in \mathbb{R}^m$ and $\bar{x} = k^{-1} \sum_j x_j$.

$$\sum_{a < b} \|x_a - x_b\|^2 = \frac{1}{2} \sum_{a,b} \|x_a - x_b\|^2.$$

Expanding,

$$\frac{1}{2} \sum_{a,b} (\|x_a\|^2 + \|x_b\|^2 - 2x_a^T x_b) = k \sum_a \|x_a\|^2 - \left\| \sum_a x_a \right\|^2.$$

But

$$\sum_a \|x_a - \bar{x}\|^2 = \sum_a \|x_a\|^2 - k \|\bar{x}\|^2.$$

Hence

$$\sum_{a < b} \|x_a - x_b\|^2 = kS^2(X).$$

Why Centroid Size?

Centroid size is useful because:

- it is simple to compute from coordinates;
- it is invariant to translation and rotation;
- it behaves linearly under scaling;
- it uses all landmarks, not just one baseline;
- it gives a natural normalization for pre-shape coordinates.

However, it is not the only possible size measure. Biological size may sometimes be better represented by mass, area, volume, or a scientifically chosen length.

Removing Scale

After centring, remove scale by normalizing:

$$Z = \frac{CX}{\|CX\|_F}.$$

Then

$$\mathbf{1}_k^T Z = 0, \quad \|Z\|_F = 1.$$

Z is called a **pre-shape**. It has translation and scale removed, but still contains orientation.

Pre-shape Space

The space of centred, unit-size configurations is

$$\mathcal{S}_m^k = \{Z \in \mathbb{R}^{k \times m} : \mathbf{1}_k^T Z = 0, \|Z\|_F = 1\}.$$

Since centring imposes m linear constraints, the centred space has dimension $m(k - 1)$.
The unit norm condition removes one more dimension:

$$\mathcal{S}_m^k \cong S^{m(k-1)-1}.$$

Pre-shape space is a sphere.

Mini-Exercise 3

Let $k = 4$ landmarks in $m = 2$ dimensions.

- ① What is the dimension of configuration space?
- ② What is the dimension after removing translation?
- ③ What is the dimension after removing translation and scale?
- ④ What is the dimension after also removing rotation?

Mini-Exercise 3

Let $k = 4$ landmarks in $m = 2$ dimensions.

- ① What is the dimension of configuration space?
- ② What is the dimension after removing translation?
- ③ What is the dimension after removing translation and scale?
- ④ What is the dimension after also removing rotation?

• **Answer:**

$$8 \rightarrow 6 \rightarrow 5 \rightarrow 4.$$

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Rotations Remain

Pre-shapes still depend on orientation:

$$Z \mapsto ZR, \quad R \in SO(m).$$

Thus a shape is an orbit:

$$[Z] = \{ZR : R \in SO(m)\}.$$

Kendall shape space is the quotient

$$\Sigma_m^k = \mathcal{S}_m^k / SO(m).$$

Key point

Taking a quotient by rotations makes the sample space curved and sometimes singular.

Dimension of Shape Space

If the action is regular, then

$$\dim(\Sigma_m^k) = \dim(\mathcal{S}_m^k) - \dim(SO(m)).$$

Since

$$\dim(\mathcal{S}_m^k) = m(k - 1) - 1$$

and

$$\dim(SO(m)) = \frac{m(m - 1)}{2},$$

we obtain

$$\dim(\Sigma_m^k) = m(k - 1) - 1 - \frac{m(m - 1)}{2}.$$

Dimension Examples

k	m	Shape-space dimension	Comment
3	2	2	triangle shapes
4	2	4	planar quadrilateral shapes
6	2	8	mouse vertebra landmarks
7	3	11	macaque skull landmarks
22	3	56	DNA phosphorus atoms

Even moderate k yields high-dimensional shape variables.

Planar Shape and Complex Coordinates

For $m = 2$, write centred configurations as complex vectors $z \in \mathbb{C}^k$ with $\mathbf{1}^* z = 0$ and $\|z\| = 1$. Rotation is multiplication by a unit complex number:

$$z \mapsto e^{i\theta} z.$$

Therefore planar Kendall shape space is

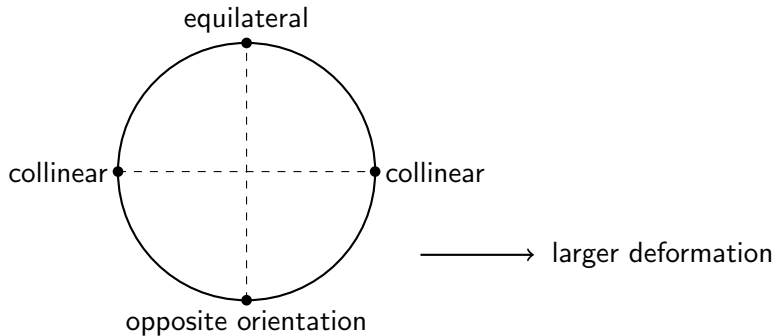
$$\Sigma_2^k \cong \mathbb{CP}^{k-2}.$$

For triangles:

$$\Sigma_2^3 \cong \mathbb{CP}^1 \cong S^2.$$

Shape Space of Triangles

Kendall sphere for labelled triangles



Triangle shape space provides a useful mental model: shape variables can live on a curved surface, not a plane.

Orbits and Horizontal Directions

At a pre-shape Z , rotations generate the **vertical space**:

$$\mathcal{V}_Z = \{ZA : A^T = -A\}.$$

Directions orthogonal to rotations form a **horizontal space**:

$$\mathcal{H}_Z = \{V : \langle V, Z \rangle = 0, Z^T V \text{ is symmetric}\}.$$

Shape variation is represented by horizontal tangent vectors, not arbitrary perturbations.

Geodesic Distance on the Pre-shape Sphere

For two pre-shapes $Z_1, Z_2 \in \mathcal{S}_m^k$, the spherical distance is

$$\rho(Z_1, Z_2) = \arccos \langle Z_1, Z_2 \rangle,$$

where

$$\langle A, B \rangle = \text{tr}(A^T B).$$

But shape distance should minimize over rotations:

$$\rho_K([Z_1], [Z_2]) = \min_{R \in SO(m)} \arccos \langle Z_1, Z_2 R \rangle.$$

Full Procrustes Distance

A chordal alternative is the full Procrustes distance:

$$d_F([Z_1], [Z_2]) = \min_{R \in SO(m), \beta > 0} \|Z_1 - \beta Z_2 R\|_F.$$

For well-aligned pre-shapes with angle θ ,

$$d_F = \sin \theta.$$

The ordinary Procrustes distance is

$$d_P([Z_1], [Z_2]) = \min_{R \in SO(m)} \|Z_1 - Z_2 R\|_F.$$

For angle θ ,

$$d_P = 2 \sin(\theta/2).$$

Local Euclidean Approximation

Most practical statistical methods use tangent coordinates near the mean shape.

For small θ :

$$\sin \theta \approx \theta, \quad 2 \sin(\theta/2) \approx \theta.$$

Hence different shape distances agree to first order near the mean.

Practical implication

If shape variability is small, Euclidean multivariate analysis of tangent coordinates can work well.

When Geometry Matters

The tangent approximation may be poor when:

- variability is large;
- the mean is close to singular shapes;
- data have multiple clusters;
- rotations are unstable;
- correspondence uncertainty is large.

Then one may need intrinsic means, geodesic regression, Bayesian manifold models, or nonparametric methods.

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Problem Setup

Given centred, scaled configurations X and Y , align Y to X by choosing a rotation:

$$\hat{R} = \arg \min_{R \in SO(m)} \|X - YR\|_F^2.$$

Expand the objective:

$$\|X - YR\|_F^2 = \|X\|_F^2 + \|Y\|_F^2 - 2\text{tr}(X^T YR).$$

For pre-shapes, $\|X\|_F = \|Y\|_F = 1$, so minimizing distance is equivalent to maximizing

$$\text{tr}(X^T YR).$$

SVD Solution

Let

$$X^T Y = U D V^T,$$

where $D = \text{diag}(d_1, \dots, d_m)$ has nonnegative singular values.

Then the optimal rotation, ignoring the determinant constraint, is

$$\hat{R} = V U^T.$$

If $\det(V U^T) = -1$ and reflections are not allowed, use

$$\hat{R} = V \begin{bmatrix} 1 & & 0 \\ & \ddots & \\ 0 & & \det(V U^T) \end{bmatrix} U^T.$$

Derivation of the SVD Solution

We need to maximize

$$\text{tr}(X^T YR) = \text{tr}(UDV^T R).$$

By cyclicity of trace,

$$\text{tr}(UDV^T R) = \text{tr}(DV^T RU).$$

Let

$$Q = V^T RU.$$

Then Q is orthogonal and

$$\text{tr}(DQ) = \sum_{j=1}^m d_j q_{jj} \leq \sum_{j=1}^m d_j.$$

Equality occurs at $Q = I$, giving $R = VU^T$.

Ordinary Procrustes Distance

At the optimum,

$$\max_{R \in SO(m)} \text{tr}(X^T Y R) = \sum_j d_j.$$

Therefore

$$d_P^2([X], [Y]) = \min_R \|X - YR\|_F^2 = 2 \left(1 - \sum_j d_j \right).$$

The singular values of $X^T Y$ summarize the optimal similarity of two pre-shapes.

Reflection Shape

Sometimes mirror images should be equivalent. Then use the full orthogonal group:

$$O(m) = \{R : R^T R = I\}.$$

The reflection Procrustes fit is

$$\min_{R \in O(m)} \|X - YR\|_F.$$

Use reflections only if they are scientifically meaningful. For example:

- left/right anatomical symmetry: perhaps yes;
- molecules with chirality: usually no;
- handwritten symbols: often no.

Partial Procrustes Fit

If scale is retained or estimated, solve

$$\min_{\beta, R} \|X - \beta YR\|_F^2.$$

For fixed R ,

$$\hat{\beta}(R) = \frac{\text{tr}(X^T YR)}{\|Y\|_F^2}.$$

At the optimal rotation,

$$\hat{\beta} = \frac{\sum_j d_j}{\|Y\|_F^2}.$$

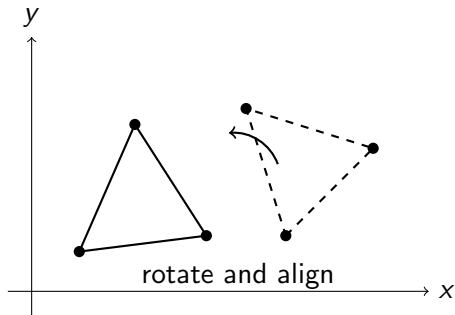
This is useful for size-and-shape analysis.

Algorithm: Pairwise Procrustes

- 1 Centre X and Y .
- 2 Scale to unit centroid size if doing shape analysis.
- 3 Compute $M = X^T Y$.
- 4 Compute SVD $M = UDV^T$.
- 5 Set $\hat{R} = VU^T$, with determinant correction if needed.
- 6 Use $Y\hat{R}$ for the aligned configuration.

This algorithm is small, stable, and central to almost everything that follows.

A Geometric View



Procrustes analysis is least-squares alignment after nuisance effects have been removed.

R: Pairwise Procrustes Fit

```
centre <- function(X) scale(X, scale = FALSE)
csize  <- function(X) sqrt(sum(centre(X)^2))
preshape <- function(X) centre(X) / csize(X)
X <- dna.dat[, , 1]
Y <- dna.dat[, , 30]
X0 <- preshape(X)
Y0 <- preshape(Y)

sv <- svd(t(X0) %*% Y0)
R  <- sv$v %*% t(sv$u)
determinant(R, logarithm=FALSE)
Y_aligned <- Y0 %*% R
dP <- sqrt(sum((X0 - Y_aligned)^2))
dP

print( sqrt( 2*(1-cos( riemdist( X,Y) ) )))    #check
```

From Two Shapes to a Sample

For n configurations $X_1, \dots, X_n$, generalized Procrustes analysis estimates a common mean shape and aligns all specimens.

A common formulation is

$$\min_{\mu, R_1, \dots, R_n} \sum_{i=1}^n \|X_i R_i - \mu\|_F^2,$$

subject to

$$\|\mu\|_F = 1, \quad \mathbf{1}^T \mu = 0.$$

GPA Algorithm

- 1 Centre and scale all configurations.
- 2 Initialize μ using one configuration or a preliminary average.
- 3 For each i , align X_i to μ by ordinary Procrustes analysis.
- 4 Update

$$\mu \leftarrow \frac{\sum_i X_i \hat{R}_i}{\left\| \sum_i X_i \hat{R}_i \right\|_F}.$$

- 5 Iterate until the change in μ is negligible.

The result is the Procrustes mean shape and registered sample.

Why GPA is Iterative

The difficulty is circular:

- to estimate the mean, we need aligned data;
- to align the data, we need a mean.

GPA alternates between registration and averaging.

This is similar in spirit to:

- EM algorithms;
- template estimation in image registration;
- Karcher mean algorithms on manifolds.

Procrustes Mean

The Procrustes mean shape can be defined extrinsically as

$$\hat{\mu} = \arg \min_{\mu \in \Sigma_m^k} \sum_{i=1}^n d_P^2([X_i], \mu).$$

For small variation, GPA gives a stable and interpretable estimate.

For large variation or nonunique alignment, mean shape estimation can become delicate.

Visualization After GPA

After GPA, one often plots:

- all aligned landmark configurations;
- the mean shape;
- residual vectors from mean to observations;
- deformation grids or arrows;
- principal component warps.

Do not skip visualization

Most errors in shape analysis are first discovered visually: wrong labels, flipped specimens, bad landmarks, outliers, or poor alignment.

R: GPA with the shapes Package

```
library(shapes)

data(mice)
# mice$x is k x m x n landmark array
gpa <- procGPA(mice$x)

# aligned configurations
aligned <- gpa$rotated
# mean shape
mean_shape <- gpa$mshape

plotshapes(gpa$rotated)
lines( mean_shape[c(1,6,2,3,4,5,1),], col=2)

#Some DNA data
data(dna.dat)
```

Case Study: Mouse Vertebrae

The mouse vertebrae example has three groups:

- Control;
- Large: selected for large body weight;
- Small: selected for small body weight.

Scientific questions:

- 1 Are the mean shapes different across groups?
- 2 Is shape associated with centroid size?
- 3 Which landmarks explain the main differences?

This is a natural example for GPA, tangent PCA, and group comparison.

Outliers and Quality Control

Check the following before inference:

- Are landmark labels consistent?
- Are some specimens mirrored or rotated incorrectly?
- Are centroid sizes biologically plausible?
- Do residual Procrustes distances identify outliers?
- Are semi-landmarks overfitting noise?

A formal analysis cannot rescue a poor landmarking protocol.

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Why Tangent Coordinates?

Shape observations lie on a nonlinear quotient space.

Classical methods like PCA, regression, MANOVA, and LDA are Euclidean.

The usual compromise:

$$\text{shape space near } \hat{\mu} \approx T_{\hat{\mu}} \Sigma_m^k.$$

Then each specimen is represented by a tangent vector.

Sphere Log Map

For intuition, consider the unit sphere. If x and μ are unit vectors and

$$\theta = \arccos(\mu^T x),$$

then the logarithm map at μ is

$$\log_{\mu}(x) = \frac{\theta}{\sin \theta} (x - \cos \theta \mu).$$

This vector lies in the tangent space:

$$\mu^T \log_{\mu}(x) = 0.$$

Projection Approximation

For small shape variation, a simpler approximation is often used:

$$v_i = X_i^* - \bar{X},$$

where X_i^* is the GPA-aligned pre-shape and $\bar{X}$ is the mean pre-shape.

Then vectorize:

$$y_i = \text{vec}(v_i) \in \mathbb{R}^{km}.$$

Finally remove redundant directions associated with translation, scaling, and rotation.

Tangent Coordinates

Let B be an orthonormal basis for the horizontal tangent space at $\hat{\mu}$.

Then tangent coordinates are

$$t_i = B^T \text{vec}(X_i^* - \hat{\mu}).$$

They have dimension

$$p = m(k-1) - 1 - \frac{m(m-1)}{2}.$$

These $t_i \in \mathbb{R}^p$ are the inputs to ordinary multivariate analysis.

Covariance in Tangent Space

The sample tangent covariance is

$$\hat{\Sigma} = \frac{1}{n-1} \sum_{i=1}^n (t_i - \bar{t})(t_i - \bar{t})^T.$$

If p is large and n is small, $\hat{\Sigma}$ is singular or unstable.

Common remedies:

- dimension reduction by PCA;
- shrinkage covariance estimation;
- permutation tests;
- regularized discriminant analysis;
- Bayesian priors.

Small-Variance Assumption

Tangent analysis is strongest when

$$\max_i d_P([X_i], \hat{\mu})$$

is small.

Diagnostics:

- compare Procrustes and tangent distances;
- plot residual distances;
- inspect whether PC scores are dominated by outliers;
- check if deformation plots remain biologically plausible.

Mini-Exercise 4: Digit 3 data in R

```
library(shapes) # load the 'shapes' package (may need installing)
digit3.dat      # Will print out the coordinates of the digit3 data
plotshapes(digit3.dat) # all 30 observations - what a mess!
plotshapes(digit3.dat, joinline=1:13) # a bit less of a mess
```

Carry out Procrustes analysis (using translation, rotation and scale)

```
out <- procGPA( digit3.dat , scale = TRUE)
out$rotated      # contains the Procrustes registered data
plotshapes ( out$rotated , joinline=c(1:13) ) # Procrustes registered data
out$mshape      # contains the Procrustes mean shape
lines( out$mshape, col=2 ) # adds the mean shape to the plot (in red)
plot( out$size , out$rho )
```

Note 'rho' is called the Riemannian distance, denote as $\rho = \arccos(1 - d_p^2/2)$. Here 'rho' is a vector of shape distances to the mean ($0 \leq \rho \leq \pi/2$).

Mini-Exercise 4 (continued)

```
plot( out$size , out$rho )  
text( out$size , out$rho , c(1:30) )
```

Is there a possible outlier? Which observation?

```
plotshapes( digit3.dat , digit3.dat[,1] , joinline= 1:13)  
out2 <- procGPA( digit3.dat[-1,,] , scale = TRUE)    #Procrustes w/o the outlier  
plot( out2$size , out2$rho )  
text( out2$size , out2$rho , c(2:30) )
```

Some further questions:

- 1 What is the dimension of planar shape space here?
- 2 If $n = 30$, is the sample covariance matrix full rank?

Mini-Exercise 4 (continued)

```
plot( out$size , out$rho )  
text( out$size , out$rho , c(1:30) )
```

Is there a possible outlier? Which observation?

```
plotshapes( digit3.dat , digit3.dat[,1] , joinline= 1:13)  
out2 <- procGPA( digit3.dat[-1,,] , scale = TRUE)    #Procrustes w/o the outlier  
plot( out2$size , out2$rho )  
text( out2$size , out2$rho , c(2:30) )
```

Some further questions:

- 1 What is the dimension of planar shape space here?
- 2 If $n = 30$, is the sample covariance matrix full rank?

• Answers:

$$p = 2k - 4 = 22,$$

so covariance rank is $\min(n - 1, p)$ so it is full rank, but regularization may still be wise.

PCA in Tangent Space

Given tangent coordinates $t_1, \dots, t_n \in \mathbb{R}^p$, compute

$$\hat{\Sigma} q_j = \lambda_j q_j.$$

PC scores:

$$s_{ij} = q_j^T (t_i - \bar{t}).$$

The first few eigenvectors describe the main modes of shape variation.

Visualizing a Principal Mode

To visualize PC j , plot shapes corresponding to

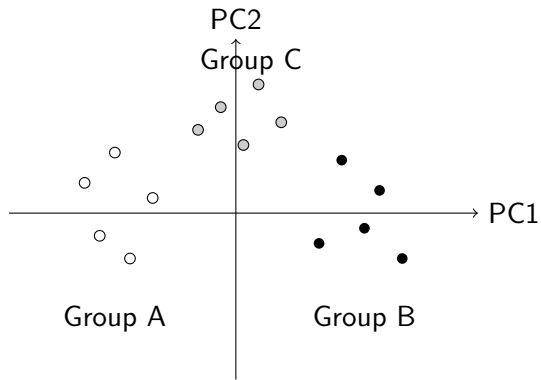
$$\bar{t} \pm c\sqrt{\lambda_j}q_j,$$

where common choices are $c = 2$ or 3 .

Then map the tangent vectors back to landmark coordinates and display:

- landmark shifts;
- deformation arrows;
- thin-plate spline grids;
- overlaid outlines.

PCA Score Plot



Score plots are exploratory; confirm group differences by inference.

Interpreting PCA Carefully

PCA components are mathematical summaries, not automatically biological mechanisms.

Ask:

- Is PC1 mostly size-related allometry that escaped shape normalization?
- Are group differences visible in PC score plots?
- Are large deformations within the observed data range?
- Do PCs correspond to localized landmark movement or global bending?

Mini-Exercise 5

PCA for Digit 3.

```
out<-procGPA( digit3 )  
shapepca( out )    # plot the effects of the first 3 PCs  
out2<-procGPA( digit3[-1,,] )      # without outlier  
shapepca( out2 )
```

Dataset Mice. Using different tangent coordinates.

```
data(mice)  
gpa <- procGPA(mice$x)  
shapepca(gpa,type="r",mag=3)  
shapepca(gpa,type="v",mag=3)  
shapepca(gpa,type="g",mag=3)  
plot(gpa$scores,col=as.factor(mice$group),pch=as.character(mice$group))  
gpa2 <- procGPA(mice$x,tangentcoords="expomap")  
plot(gpa2$scores,col=as.factor(mice$group),pch=as.character(mice$group))
```

Any difference in the choice of tangent coordinates?

Outline

- 1 Motivation and Data Structures
- 2 Removing Translation and Measuring Size
- 3 Shape Spaces and Geometry
- 4 Procrustes Analysis
- 5 Tangent Coordinates and Multivariate Statistics
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- 7 Outlines, Curves, Surfaces, and Unlabelled Data
- 8 Modern Directions
- 9 Summary and Discussion

High-Dimension, Low-Sample-Size

Shape problems often have many landmarks and few specimens.

If $p \gg n$, then:

- sample covariance is singular;
- PCs after rank $n - 1$ are undefined;
- group separation may be overfit;
- cross-validation becomes essential.

Use dimension reduction, permutation inference, and careful validation.

Two-Sample Shape Problem

Suppose two groups have mean shapes μ_1 and μ_2 .

The null hypothesis is

$$H_0 : \mu_1 = \mu_2.$$

In tangent coordinates, a simple test statistic is

$$T = \|\bar{t}_1 - \bar{t}_2\|^2.$$

But the null distribution is not always standard, especially when p is large.

Permutation Testing

Permutation tests are natural because labels are exchangeable under H_0 .

Algorithm:

- 1 Compute observed statistic T_{obs} .
- 2 Randomly permute group labels.
- 3 Recompute T_b for each permutation b .
- 4 Estimate

$$p = \frac{1 + \#\{b : T_b \geq T_{obs}\}}{1 + B}.$$

Use the same GPA/tangent pipeline inside each permutation if alignment depends strongly on groups.

MANOVA in Tangent Space

For G groups, a multivariate model is

$$t_i = \alpha_{g(i)} + \varepsilon_i.$$

Classical statistics include:

- Wilks' lambda;
- Pillai trace;
- Hotelling-Lawley trace;
- Roy's largest root.

In shape analysis, these are often paired with permutation calibration.

Procrustes ANOVA

A Procrustes ANOVA partitions squared Procrustes variation:

$$SS_{total} = SS_{between} + SS_{within}.$$

The analysis resembles ordinary ANOVA but uses aligned landmark coordinates or tangent coordinates.

Useful for:

- group effects;
- individual effects in repeated measures;
- fluctuating asymmetry;
- measurement error designs.

Allometry

Allometry is shape change associated with size.

A common tangent-space model is

$$t_i = \alpha + \beta \log S_i + \varepsilon_i,$$

where S_i is centroid size.

For groups:

$$t_i = \alpha_{g(i)} + \beta_{g(i)} \log S_i + \varepsilon_i.$$

Questions:

- Is shape related to size?
- Do groups have different allometric slopes?

Regression with Covariates

More generally,

$$t_i = B^T x_i + \varepsilon_i,$$

where x_i includes age, treatment, sex, disease status, genotype, or environmental variables.

For longitudinal data:

$$t_{ij} = f(t_{ij}^{time}) + b_i + \varepsilon_{ij}.$$

Care is needed because repeated shapes from the same individual are correlated.

Classification

Classification pipeline:

- ① GPA on training data.
- ② Project to tangent coordinates.
- ③ Reduce dimension, often by PCA.
- ④ Fit classifier: LDA, QDA, logistic regression, SVM, random forest.
- ⑤ Cross-validate the full pipeline.

Never select PCs or landmarks using all data before cross-validation; this leaks test information.

Effect Size

A statistically significant shape difference may be small.

Report:

- Procrustes distance between mean shapes;
- percentage of variation explained by group;
- deformation plot of group difference;
- confidence regions or bootstrap uncertainty;
- classification accuracy only with validation.

Mini-Exercise 6

Design a permutation test for the mouse vertebrae groups.
Specify:

- 1 the null hypothesis;
- 2 the test statistic;
- 3 how labels are permuted;
- 4 how many permutations you would use;
- 5 how you would visualize the effect if significant.

Growth Studies

Growth data measure the same organism at multiple time points.

Examples:

- rat skulls measured at several ages;
- human movement trajectories;
- developmental biology time courses;
- disease progression in medical imaging.

Key issue: observations within an individual are correlated.

Linear Shape Regression

For scalar covariate x_i :

$$t_i = \alpha + \beta x_i + \varepsilon_i.$$

The fitted shape trend from $x = a$ to $x = b$ is

$$\Delta t = \hat{\beta}(b - a).$$

Visualize by mapping

$$\hat{\mu} \pm c\Delta t$$

back to landmark configurations.

Polynomial and Spline Trends

For nonlinear growth:

$$t_i = \alpha + \beta_1 x_i + \beta_2 x_i^2 + \varepsilon_i,$$

or

$$t_i = \sum_{r=1}^R \beta_r B_r(x_i) + \varepsilon_i,$$

where B_r are spline basis functions.

Use smoothness only when the temporal sampling supports it.

Mixed-Effects Shape Models

For individual i at time j :

$$t_{ij} = \alpha + f(a_{ij}) + b_i + \varepsilon_{ij}.$$

Here:

- a_{ij} is age or time;
- b_i is an individual random effect;
- ε_{ij} is residual shape variation.

This separates population growth from subject-specific variation.

Geodesic Regression

A more intrinsic model fits a geodesic in shape space:

$$\mu(t) = \text{Exp}_{\mu_0}(tv).$$

Estimate μ_0 and tangent velocity v by minimizing

$$\sum_i d^2([X_i], \mu(t_i)).$$

This is geometrically elegant but computationally more demanding.

Allometry vs Growth

Growth usually changes both size and shape.

Possible models:

$$\log S_i = a_0 + a_1 \text{ age}_i + e_i,$$

$$t_i = b_0 + b_1 \text{ age}_i + b_2 \log S_i + \varepsilon_i.$$

This helps separate:

- age-related shape change;
- size-related shape change;
- residual developmental variation.

Mini-Exercise 7

A longitudinal skull dataset has 18 animals measured at 8 ages.
What are the risks of treating all 18×8 shapes as independent?

Expected answer:

- standard errors too small;
- inflated type I error;
- individual differences confounded with growth;
- misleading PCA if subject trajectories dominate.

Practical Workflow

- 1 Define the scientific question.
- 2 Choose landmarks or curves and justify correspondence.
- 3 Perform quality control and visualization.
- 4 Compute centroid sizes and pre-shapes.
- 5 Run GPA.
- 6 Check residuals and outliers.
- 7 Compute tangent coordinates.
- 8 Explore with PCA.
- 9 Fit inferential models.
- 10 Visualize effects as deformations.

Useful R packages:

- `shapes`: classical statistical shape analysis;
- `geomorph`: geometric morphometrics workflows;
- `Morpho`: 3D morphometrics and meshes;
- `rgl`: 3D visualization.

Outline

- 1 Motivation and Data Structures
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- 9 Summary and Discussion

Beyond Landmarks

Many objects are better represented as:

- closed curves;
- open curves;
- surfaces;
- images;
- volumes;
- point clouds.

Landmark methods remain important, but correspondence and parameterization become central.

Pseudo-Landmarks

A simple way to handle outlines is to sample many points along the curve.

Advantages:

- easy to combine with landmark methods;
- works with GPA and PCA;
- simple visualization.

Disadvantages:

- correspondence can be arbitrary;
- equally spaced points may not match biological features;
- high-dimensional covariance problems appear quickly.

Semi-Landmarks

Semi-landmarks can slide along curves or surfaces to improve correspondence.

Typical criteria:

- minimize bending energy;
- minimize Procrustes distance;
- preserve smoothness along the curve.

This introduces an additional optimization step and must be reported clearly.

Elastic Shape Analysis

For parameterized curves $\beta(t)$, elastic methods compare shapes while allowing reparameterization:

$$\beta(t) \mapsto \beta(\gamma(t)),$$

where γ is a warping function.

The goal is invariance to:

- translation;
- rotation;
- scale;
- parameterization.

Square-root velocity framework:

$$q(t) = \frac{\dot{\beta}(t)}{\sqrt{\|\dot{\beta}(t)\|}}.$$

Distances become geodesic lengths in elastic shape spaces.

Extensions to surfaces and reparameterization-invariant geometry.

Surface Shape

Surface data arise in neuroimaging, anatomy, molecular science, and graphics.

Challenges:

- dense high-dimensional data;
- topology and mesh quality;
- registration uncertainty;
- local vs global deformation;
- multiple comparisons over vertices.

Methods include currents, varifolds, diffeomorphic registration, and surface PCA.

Unlabelled Point Sets

When labels are unknown, estimate both transformation and correspondence:

$$\min_{P,R,\beta,\gamma} \|X - \beta PYR - \mathbf{1}\gamma^T\|^2.$$

This appears in:

- protein active-site matching;
- object recognition;
- spatial point patterns;
- 3D scans and point clouds.

The problem is often multimodal.

Bayesian Matching

A Bayesian formulation treats correspondence as missing data:

$$p(P, R, \theta \mid X, Y) \propto p(X, Y \mid P, R, \theta)p(P)p(R)p(\theta).$$

Benefits:

- uncertainty in matching is propagated;
- prior information can be included;
- multiple plausible alignments are retained.

Cost: computation is often much heavier.

Mini-Exercise 8

For a closed outline with no natural starting point, why is ordinary labelled Procrustes analysis problematic?

Answer?

Mini-Exercise 8

For a closed outline with no natural starting point, why is ordinary labelled Procrustes analysis problematic?

Answer?

- point 1 on one curve may not correspond to point 1 on another;
- cyclic shifts can change the apparent shape distance;
- parameterization, not just geometry, affects the result.

Common Mistakes

Avoid these

- treating raw coordinates as shape variables;
- ignoring size when allometry is central;
- using too many pseudo-landmarks with too few specimens;
- interpreting PCs without deformation plots;
- doing feature selection before cross-validation;
- forgetting correspondence uncertainty;
- reporting only p-values, not effect sizes.

Outline

- 1 Motivation and Data Structures
- 2 Removing Translation and Measuring Size
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- 4 Procrustes Analysis
- 5 Tangent Coordinates and Multivariate Statistics
- 6 Inference and Group Comparisons
- 7 Outlines, Curves, Surfaces, and Unlabelled Data
- 8 Modern Directions**
- 9 Summary and Discussion

Principal Nested Shape Spaces (PNS)

Classical tangent PCA linearizes shape space locally:

$$\Sigma_m^k \approx T_\mu(\Sigma_m^k).$$

PNS provides a nonlinear alternative based on nested subspheres.

- Developed by Jung, Dryden and Marron
- Designed for spherical and shape data
- Captures curved variability efficiently
- Important extension of nonlinear dimension reduction

Nested Subspheres

For data on a sphere:

$$x_i \in S^d,$$

PNS recursively fits:

$$S^d \supset S^{d-1} \supset \dots \supset S^1.$$

At each stage:

- ① fit a lower-dimensional subsphere;
- ② project the data;
- ③ continue recursively.

This differs fundamentally from linear PCA.

Great Spheres and Small Spheres

Great spheres are geodesic submanifolds.

Small spheres:

$$S^{d-1}(c, r) = \{x : d(x, c) = r\}$$

allow strongly curved modes of variation.

Applications:

- growth trajectories,
- articulated motion,
- protein conformations,
- nonlinear biological variability.

PNSS for Shape Analysis

After Procrustes alignment:

$$Z_i \in S^{m(k-1)-1}.$$

PNSS operates directly on the pre-shape sphere.

Advantages:

- lower reconstruction error;
- interpretable nonlinear coordinates;
- fewer dimensions often required;
- geometrically natural representation.

PNS and Modern Manifold Learning

PNS connects naturally with:

- manifold learning;
- nonlinear dimension reduction;
- geometric statistics;
- latent manifold models;
- geometric deep learning.

Curved data should be modelled using curved geometry.

Mini-Exercise 9

Carry out Procrustes analysis, PCA and PNSS of the monkey skull data.

```
tem <- read.csv("monkey.csv")
tem<-as.matrix(tem)
dim(tem) #Note the first column has a landmark label (number) in it.
tem1 <- tem[,2:58]
monk <- array(tem1, c(7, 3, 19) )
```

Carry out Procrustes and tangent PCA

```
library(shapes)
ans <- procGPA( monk)
ans$percent
plot(ans$rho)
shapepca( ans )
```

Mini-Exercise 9 (continued)

Carry out PNSS with small sphere decompositions. Is the percentage of variability explained similar to PCA?

```
ans2 <- pnss3d( monk , n.pc=10, sphere.type="small")  
ans2$percent  
out <- plot3darcs( ans2, pcno=1, c=3)
```

There seems to be an outlier. Let's remove it and carry out the analysis again.

```
ans3 <- pnss3d( monk[,,-19] , n.pc=10, sphere.type="small")  
out <- plot3darcs( ans3, pcno=1, c=3)
```

Are the PCA and PNSS similar to before or not? What happens if we use great sphere decompositions (sphere.type="great")

Wasserstein Geometry

Optimal transport distance:

$$W_2^2(\mu, \nu) = \inf_{\pi \in \Pi(\mu, \nu)} \int \|x - y\|^2 d\pi(x, y).$$

Applications:

- shape distributions,
- registration,
- generative modeling,
- computational anatomy.

LDDMM: Large Deformation Diffeomorphic Metric Mapping

LDDMM is a foundational framework in computational anatomy. Main idea:

$$\phi_t : \Omega \rightarrow \Omega$$

is a smooth time-dependent diffeomorphism evolving according to:

$$\frac{d\phi_t}{dt} = v_t \circ \phi_t,$$

where:

$$v_t \in V$$

belongs to a reproducing kernel Hilbert space of vector fields.

Optimization problem:

$$E(v) = \int_0^1 \|v_t\|_V^2 dt + D(I_0 \circ \phi_1^{-1}, I_1),$$

balancing:

- deformation regularity,
- image matching fidelity.

Geometry and Applications of LDDMM

LDDMM induces a Riemannian metric on shape spaces. Key consequences:

- geodesic shape evolution,
- invertible deformations,
- population anatomical statistics,
- longitudinal disease modeling.

Major applications:

- brain imaging,
- cortical registration,
- atlas construction,
- growth and aging studies.

Foundational contributors:

- Trounev, Younes, Miller, Beg.

Deep Shape Models

Current frontier:

Deep Learning + Geometry.

Examples:

- geometric autoencoders,
- graph neural networks,
- manifold diffusion models,
- neural implicit surfaces.

Modern shape analysis increasingly combines:

Statistics + Geometry + AI

Outline

- 1 Motivation and Data Structures
- 2 Removing Translation and Measuring Size
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- 9 Summary and Discussion**

Conceptual Summary

- Shape is geometry modulo translation, rotation, and scale.
- Landmark configurations become pre-shapes after centring and scaling.
- Kendall shape space is a quotient of the pre-shape sphere by rotations.
- Procrustes analysis solves registration using the SVD.
- GPA estimates a mean shape and aligned sample.
- Tangent coordinates make multivariate statistics possible.
- Visualization is essential for interpretation.

What to Remember

One-sentence version

Statistical shape analysis is multivariate statistics after carefully quotienting out nuisance geometry.

Practical version

Most applied projects reduce to: landmark carefully, align by GPA, inspect, project to tangent space, model, and visualize deformations.

Research version

The frontier begins where tangent approximations, fixed correspondences, or small-variance assumptions break down.

Suggested Reading

- Dryden, I. L. and Mardia, K. V. (2016). *Statistical Shape Analysis, with Applications in R*. Wiley.
- Kendall, D. G., Barden, D., Carne, T. K. and Le, H. (1999). *Shape and Shape Theory*. Wiley.
- Bookstein, F. L. (1991). *Morphometric Tools for Landmark Data*. Cambridge University Press.
- Srivastava, A. and Klassen, E. (2016). *Functional and Shape Data Analysis*. Springer.

Final Discussion

Questions to take away:

- ① What is the correct invariance group for your objects?
- ② Is correspondence known, estimated, or scientifically ambiguous?
- ③ Is tangent-space analysis adequate?
- ④ What visualizations make the fitted effects interpretable?
- ⑤ What uncertainty should be propagated?

Thank you.